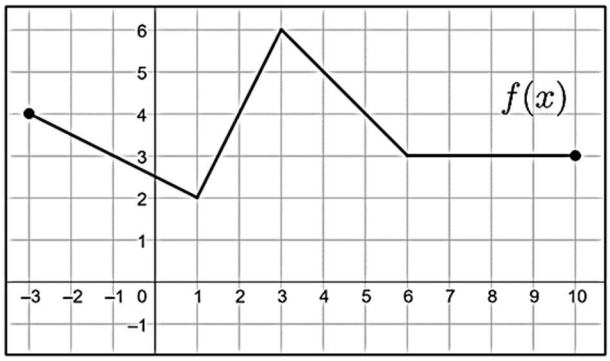
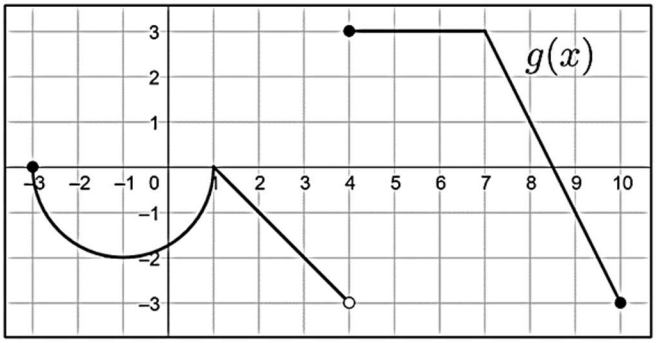
Worksheet 3-5 & 3-6 AB
Derivative Procedures and Higher Order Derivatives #2
AP Calculus Name_____

Directions: The graphs of f and g are below along with selected values of the twice differentiable function h and its derivative h'. The function $k(x)$ is implicitly defined algebraically





x	-2	-1	1	4	7	9
$h(x)$	3	-2	4	0	-3	5
$h'(x)$	2	0	-3	7	6	-1

$$3x - \left(k(x)\right)^2 = f(x) - 2 \text{ for } k(x) \geq 0$$

1. Let $\dfrac{dy}{dx} = g\big(f(x)\big)$. Find $\dfrac{d^2y}{dx^2}$ at $x = 5$.

2. Find $k'(4)$.

3. Let $c(x) = \dfrac{g(x)}{h(x)}$. Find $c'(9)$.

4. Let $d(x) = \dfrac{1}{\sqrt{x}} + 3h(x)$. Find $d'(4)$.

You need to do these on notebook paper.

5.) Given $\cos y = 2x^2$, find $\frac{d^2y}{dx^2}$

6.) If $\frac{dy}{dx} = e^y + x^{3/2}$, find $\frac{d^2y}{dx^2}$ at $(4, 0)$

7.) If $x^2 - 2xy + 3y^2 = 8$, then $\frac{dy}{dx} =$

(A) $\frac{8+2y-2x}{6y-2x}$ (B) $\frac{3y-x}{y-x}$ (C) $\frac{2x-2y}{6y-2x}$ (D) $\frac{1}{3}$ (E) $\frac{y-x}{3y-x}$

8.) If $f(x) = \cos(e^{-x})$, then $f'(x)$ is

(A) $-\sin(e^{-x})$ (B) $\sin(e^{-x})$ (C) $\sin(e^{-x}) - e^{-x}$

(D) $\mathrm{e}^{-x}\sin(e^{-x})$ (E) $-\mathrm{e}^{-x}\sin(e^{-x})$

9.) A curve given by the equation $x^3 + xy = 8$ has slope given by $\frac{dy}{dx} = \frac{-3x^2-y}{x}$. The value of $\frac{d^2y}{dx^2}$ at the point where $x = 2$ is

(A) –6 (B) –3 (C) 0 (D) 4 (E) undefined

10.) Let f be the function defined by $f(x) = 2x^4 - x^3 + x$. Let $g(x) = f^{-1}(x)$, where $g(2) = 1$. What is the value of $g'(2)$?

(A) $\frac{1}{53}$ (B) $\frac{1}{26}$ (C) $\frac{1}{6}$ (D) $\frac{1}{2}$ (E) 53